George Abu-Aqil, Ph.D.

Post-Doctoral Researcher – Boston University

Boston, MA, USA
Gabuaqil@bu.edu, George.amg65@gmail.com
+1 (857) 972 4307

Research Profile

Multidisciplinary biomedical researcher integrating microbiology, vibrational spectroscopy, and machine learning to develop rapid diagnostic technologies. Expertise in single-cell analysis, antimicrobial resistance, and clinical translation of spectroscopy-based methods. Demonstrated track record of high-impact publications, international conference presentations, and cross-disciplinary collaborations bridging biology, engineering, and data science.

Education

2019-2023

Ph.D. degree in Medical Science

Department of Microbiology, Immunology and Genetics, Faculty of Health Science, Ben-Gurion University of the Negev in Combined Doctoral Track.

- Awards: Mid-way Negev–Tsin Scholarship; Dean's Excellence Prize.
- Thesis and publications focused on spectroscopy-driven diagnostics with ML.

2017-2019

M.Sc. degree in Medical Science

Department of Microbiology, Immunology and Genetics, Ben-Gurion University of the Negev.

2014-2017

B.Sc. degree in Medical Laboratory Science,

Faculty of Health Science, Ben-Gurion University of the Negev.

Core Skills & Expertise

- **Experimental & Analytical**. Applied microbiology (clinical pathogens, AST), FTIR/Raman & photothermal spectroscopy, qPCR/RT-PCR, hyperspectral imaging.
- **Computational & Data**. Machine learning, multivariate analysis, spectral processing, high-dimensional biological data (Python, MATLAB).
- **Research & Translation**. Experimental design, method development, integration of microbiology–spectroscopy–ML for clinical applications.
- **Leadership & Collaboration**. CTO experience; Mentoring researchers; cross-functional work with chemists, engineers, physicians, and clinicians; technical reporting, scientific writing, and presentations.

Professional experience

2024-Present

Postdoctoral Researcher, Optical Imaging & Spectroscopy

Photonic Center, Department of Electrical and Computer Engineering, Boston University.

- Investigating microbial metabolic signatures and single-cell phenotyping using advanced spectroscopy systems (Mid-IR photothermal & fluorescence imaging).
- Designing multidisciplinary experiments integrating microbiology, optics, and computational analysis.
- Training graduate researchers and establishing laboratory workflows.
- Preparing publications, grant materials, and technical reports.

2023-2024

Postdoctoral Researcher, Medical Science

Department of Microbiology, Immunology and Genetics, Ben-Gurion University of the Negev.

- Led translational projects developing microbial detection and antibiotic susceptibility assays.
- Conducted studies on bacterial & fungal clinical isolates
- Supervised students and maintained laboratory SOPs and safety protocols. Collaborated with clinicians & data scientists for large-scale studies.

2022-2023

CTO, bFAST.ai

Startup, Beer-Sheva

- Directed technology development for rapid microbial analysis & diagnostic automation.
- Built experimental validation pipelines & coordinated with academic and industry partners.
- Translated scientific outputs into scalable product workflows.

2021-2023

Shift manager, Electra – Tel-Aviv University COVID-19 Laboratory

Omer

- Supervised RT-PCR workflows; ensured quality, and turnaround time.
- Maintained SOPs, QA compliance, and biosafety protocols; trained staff.

Awards and Prizes

- Mid-way Negev - Tsin scholarship 2020-2023
- Dean Excellence Prize 2022/2023
- Best Presentation award at MLE 2022 Conference
- Faculty Scholarships: 2017-2019 & 2019-2023

Publications

First Author or Contributed Equally as First Author Publications:

1. **Abu-Aqil, G.,** Dong, D., Yin, J., Ao, J., He, H., Ding, G., Xia, Q., & Cheng, J. X. (2025). Bond-Selective Imaging via Vibrational Relaxation Encoded Fluorescence. *The Journal of Physical Chemistry Letters*, *16*(48), 12401-12410.
2. **Abu-Aqil, G.,** Adawi, S., & Huleihel, M. (2024). Early and swift identification of fungal-infection using infrared spectroscopy. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, *325*, 125101.
3. **Abu-Aqil, G.**, Suleiman, M., Lapidot, I., Huleihel, M., & Salman, A. (2024). Infrared spectroscopy-based machine learning algorithms for rapid detection of Klebsiella pneumoniae isolated directly from patients' urine and determining its susceptibility to antibiotics. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, *314*, 124141.
4. **Abu-Aqil, G.**, Lapidot, I., Salman, A., & Huleihel, M. (2023). Quick Detection of Proteus and Pseudomonas in Patients' Urine and Assessing Their Antibiotic Susceptibility Using Infrared Spectroscopy and Machine Learning. *Sensors*, *23*(19), 8132.
5. **Abu-Aqil, G.**, Suleiman, M., Sharaha, U., Nesher, L., Lapidot, I., Salman, A., & Huleihel, M. (2023). Detection of extended-spectrum β-lactamase-producing bacteria isolated directly from urine by infrared spectroscopy and machine learning. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, *295*, 122634.
6. **Abu-Aqil, G.**, Suleiman, M., Sharaha, U., Lapidot, I., Huleihel, M., & Salman, A. (2023). Instant detection of extended-spectrum β-lactamase-producing bacteria from the urine of patients using infrared spectroscopy combined with machine learning. *Analyst*, *148*(5), 1130-1140.
7. **Abu-Aqil, G.**, Suleiman, M., Sharaha, U., Riesenberg, K., Lapidot, I., Huleihel, M., & Salman, A. (2023). Fast identification and susceptibility determination of E. coli isolated directly from patients' urine using infrared-spectroscopy and machine learning. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, *285*, 121909.
8. Sharaha, U.[#], **Abu-Aqil, G.[#],** Suleiman, M.[#], Riesenberg, K., Lapidot, I., Huleihel, M., & Salman, A. (2023). Rapid determination of Proteus mirabilis susceptibility to antibiotics using infrared spectroscopy in tandem with random forest. *Journal of Biophotonics*, *16*(2), e202200198.
9. **Abu-Aqil G.**, Sharaha, U., Suleiman, M., Riesenberg, K., Lapidot, I., Salman, A., & Huleihel, M. (2022). Culture-independent susceptibility determination of E. coli isolated directly from patients' urine using FTIR and machine-learning. *Analyst*, *147*(21), 4815-4823.
10. Suleiman, M.[#], **Abu-Aqil, G.[#],** Sharaha, U.[#], Riesenberg, K., Lapidot, I., Salman, A., & Huleihel, M. (2022). Infra-red spectroscopy combined with machine learning algorithms enables early determination of Pseudomonas aeruginosa's susceptibility to antibiotics. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, *274*, 121080.
11. Sharaha, U.[#], Suleiman, M.[#], **Abu-Aqil, G.[#]**, Riesenbergb, K., Lapidot, I., Salman, A., & Huleihel, M. (2021). Rapid detection of Klebsiella pneumoniae producing extended spectrum β lactamase enzymes by infrared microspectroscopy and machine learning algorithms. *Analyst*, *146*(4), 1421-1429.

12. Suleiman, M.[#], **Abu-Aqil, G.**[#], Sharaha, U.[#], Riesenbergb, K., Sagic, O., Lapidot, I., Huleihel, M., & Salman, A. (2021). Rapid detection of Klebsiella pneumoniae producing extended spectrum β lactamase enzymes by infrared microspectroscopy and machine learning algorithms. *Analyst*, *146*(4), 1421-1429.
13. **Abu-Aqil, G.,** Tsror, L., Shufan, E., Adawi, S., Mordechai, S., Huleihel, M., & Salman, A. (2020). Differentiation of Pectobacterium and Dickeya spp. phytopathogens using infrared spectroscopy and machine learning analysis. *Journal of biophotonics*, *13*(5), e201960156.

Co-Authored Publications:

1. Suleiman, M., **Abu-Aqil, G.,** Lapidot, I., Huleihel, M., & Salman, A. (2024). Significant reduction of the culturing time required for bacterial identification and antibiotic susceptibility determination by infrared spectroscopy. *Analytical Methods*.

Conferences

Invited Speaker:

1. **MedOptics 2023** – 2nd International Conference on Medical Optics. Oct. 11-17, 2023, Oludeniz, Turkey.

Oral/Poster Presentation:

1. **2026 AI4ID –** Bridging Infection and Artificial Intelligence, Jan. 28, 2026, Cambridge, Massachusetts, USA. *(Poster Presentation)*
2. **2025 Chemical Imaging GRC -** Gordon Research Conference on Chemical Imaging, Jul. 27-Aug. 1, 2025, Easton, Massachusetts, USA. *(Poster Presentation)*
3. **2025 Chemical Imaging Workshop** – Chemical Imaging Workshop, Prof. Ji-Xin Cheng Group. Jun. 17-18, 2025, Boston, Massachusetts, USA. *(Poster Presentation)*
4. **SPEC 2024** – 13th International Conference on Clinical Spectroscopy. Jun. 2-6, 2024, Ioannina, Greece. *(Poster Presentation)*
5. **ICAVS 2023** – 12th International Conference on Advanced Vibrational Spectroscopy. Aug. 27-Sep. 1, 2023, Krakow, Poland. *(Oral Presentation)*
6. **INTERM 2023** – 10th International Congress on Microscopy & Spectroscopy. Apr. 13-19, 2023, Oludeniz, Turkey. *(Poster Presentation)*
7. **Retreat 2023** – The Retreat of the Biomedical Research School. Jan. 22-23, 2023, Ma'ale HaHamisha, Israel. *(Oral Presentation)*
8. **ECSBM 2022** - 19th European Conference on Spectroscopy of Biological Molecules. Aug. 29-Sep.1, 2022, Reims, France. *(Poster Presentation)*
9. **ISM 2022** – The 55th Annual Meeting of the Israel Society for Microbiology. Jul. 4-5, 2022, Beer Sheva, Israel. *(Poster Presentation)*
10. **SPEC 2022** – 12th International Conference on Clinical Spectroscopy. Jun. 18-23, 2022, Dublin, Ireland. *(Oral Presentation)*
11. **MLE 2022** – Machine Learning in Engineering**.** Apr. 26th, 2022, Beer-Sheva, Israel. *(Oral Presentation)* **Best presentation award.**
12. **ECSBM 2019** – 18th European Conference on Spectroscopy of Biological Molecules. Aug. 19-22, 2019, Dublin, Ireland. *(Poster Presentation)*